SPE-229960-MS

From Laboratory to Engineering and Accelerated Implementation of Polymer Flooding: A Case Study of the Uzen Oil Field in Kazakhstan

M. S. Sagyndikov, Institute of Polymer Materials and Technology, Almaty, Kazakhstan / LLP Researching and Development, Kazakhstan; R. M. Kushekov, Nazarbayev University, Astana, Kazakhstan; Z. M. Alpysbay, JSC KazNIPImunaygas, Aktau, Kazakhstan; D. A. Shyrakbayev, JSC Ozenmunaygas, Zhanaozen, Kazakhstan

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Polymer flooding is the most successful technology for enhanced oil recovery (EOR), with a history of application spanning over 60 years worldwide. The simplicity of the technology, the development of the polymer production market, improvements in technological approaches, and high economic efficiency have enabled the implementation of polymer flooding across a wide range of geological and physical conditions. Thus, this paper examines the accelerated implementation of polymer flooding at the Uzen oil field in Kazakhstan. Our previous studies have investigated the relevance and effectiveness of polymer flooding for recovering residual mobile reserves trapped in reservoirs due to increased mobility ratios and permeability contrasts. The rationale for polymer flooding compared to other chemical flooding techniques has also been established. In this work, we outline further steps for the accelerated field implementation of polymer flooding.

The methodology comprises field data analysis, laboratory screening of effective polymers, water chemical composition, experimental evaluation of polymer retention in porous media, and simulation studies integrated with rheology to set polymer viscosity targets. Laboratory optimization minimized test redundancies, accelerating the identification of best-performing polymers. The sequence of experiments was streamlined by focusing exclusively on samples most suitable for specific reservoir conditions, resulting in concise recommendations for testing methodologies tailored to unique field scenarios.

In this work, we outline further steps for the accelerated field implementation of polymer flooding. To this end, a comprehensive geological and field analysis was conducted to select a pilot site, laboratory studies for screening effective polymers and choosing the optimal water source were performed, an experimental evaluation of the dynamic retention of the chosen polymer in the porous medium was carried out, and simulation studies were combined with rheological studies to determine the target viscosity of the polymer. A research plan for monitoring the process and an injection program considering specific field conditions have been developed, and finally, the feasibility of the project has been assessed. The aforementioned comprehensive work is succinctly integrated into a document titled "Project Passport," which can serve as a guide for implementing polymer flooding in other oil fields.

The introduction of polymer flooding to the Uzen oil field marks a significant advancement in Kazakhstan's EOR practices. This study presents an innovative framework, distinct from conventional approaches, that offers expedited strategies for future implementation. Recommendations emphasize the integration of novel methodologies and engineering workflows to enhance efficiency, cost-effectiveness, and scalability.

Introduction

Polymer flooding is one of the most mature and widely applied enhanced oil recovery techniques, with over six decades of global field experience. Its success lies in its relative simplicity, adaptability to a wide range of geological conditions, and the economic advantages it offers over more complex chemical flooding methods such as alkali-surfactant-polymer (ASP) flooding. Over the years, advancements in polymer chemistry, improved understanding of polymer-reservoir interactions, and refinement of field implementation strategies have expanded the applicability of this method to increasingly challenging reservoir environments.

The Uzen oil field in Kazakhstan is a mature brownfield with more than 60 years of production history, characterized by high heterogeneity, elevated formation temperature, and high brine salinity. These conditions pose challenges for polymer stability and injectivity, requiring careful selection of polymer type, water source, and operational parameters. Previous studies have established the technical justification for polymer flooding at Uzen, identifying that its geological and petrophysical properties fall within acceptable screening criteria, while noting the need for polymers with enhanced thermal and salinity tolerance.

Building on earlier laboratory screening, rheological characterization, and dynamic adsorption studies, this work presents an integrated workflow for the accelerated implementation of polymer flooding at Uzen. The approach covers pilot site selection based on geological screening and streamline simulation, optimal polymer and injection water selection, numerical modeling for viscosity targeting, and development of a detailed injection and surveillance program. The methodology emphasizes accelerated decision-making by reducing laboratory redundancy, focusing on the most representative samples, and tailoring recommendations to field-specific conditions.

In our previous work (Imanbayev et al., 2022), we outlined the importance and possibility of polymer flooding implementation in Uzen field conditions. Moreover, preliminary feasibility studies considering pilot area was done in previous work. This paper describes the complete process – from laboratory testing and engineering design to economic assessment and field surveillance planning – for a polymer flooding pilot at Uzen, providing a replicable framework for future EOR deployments in similar mature and heterogeneous reservoirs.

Uzen oilfield

Uzen is a mature brownfield with over 60 years of production history. Its substantial oil reserves are contained within Jurassic layered formations with a total thickness of 400 meters. Within the productive sequence, detailed correlation has delineated six layers characterized by lateral continuity (Fig. 1). Despite this, the high heterogeneity of this sequence is caused by the complex spatial distribution of the reservoir rock. Based on routine core analysis (RCAL), the formation's permeability varies widely, ranging from 1.3 mD to over 120 mD. This high permeability contrast (>4) is attributed to the interbedding of sandstone with impermeable layers of clay and argillite. The formation is characterized by an average porosity of 0.25, a net-to-gross ratio of 0.5, and an initial oil saturation of 0.62. Under reservoir conditions, the oil exhibits a density of 0.793 g/cm^3, a viscosity of 4.5 cP, a formation volume factor of 1.15, and a formation temperature of 62°C. Furthermore, the crude oil is characterized by high concentrations of asphaltenes (13 wt.%) and paraffin (20 wt.%), which pose significant flow assurance challenges during production (Imanbayev et al., 2022).

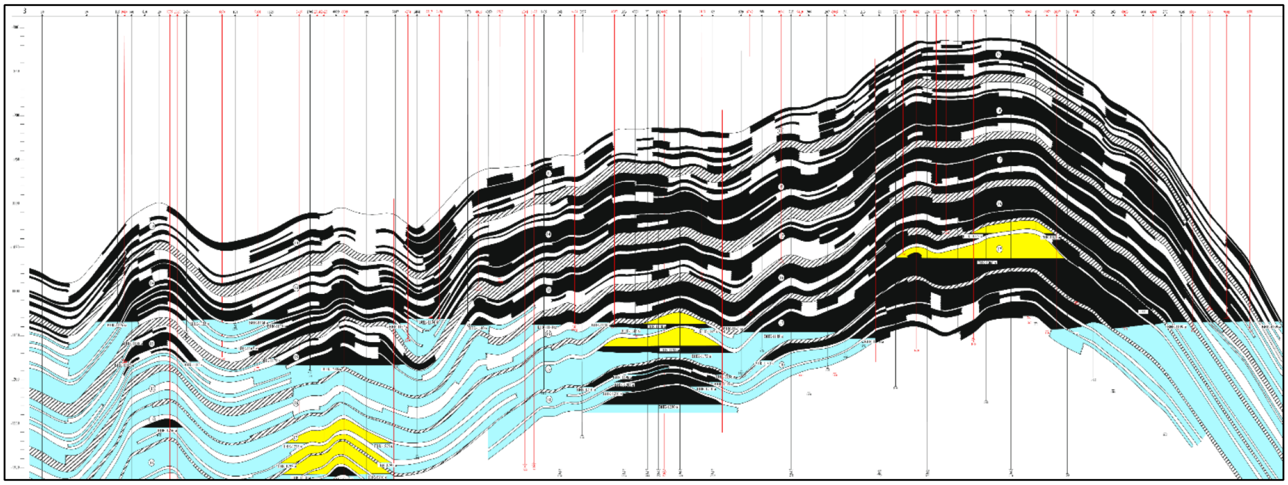

Figure 1—Geological profile of the Uzen field.

The field reached its peak production in 1975, yielding approximately 15 million tons of oil (~110 MMBBL) that year. Since then, production has been in steady decline. To mitigate this, various recovery techniques have been implemented, including hundreds of hydraulic fracturing and refracturing, conformance control in injection wells, and water shut-off treatments in producers, currently sustaining an annual output of around 5 million tons of oil (~37 MMBBL). However, this output is accompanied by roughly 45 million tons of water, resulting in a water cut exceeding 90% – a clear indication of severe water encroachment and advanced reservoir depletion. Currently, reservoir pressure is maintained through water injection via approximately 1 000 injectors, while the active production well stock comprises around 4 000 wells ("KMG Engineering" LLP, 2022).

Justification of the polymer flooding applicability

In our earlier study (Sagyndikov, Kushekov, et al., 2022) we discussed critical factors influencing polymer flooding performance by analyzing recent global field cases (Fig. 2) alongside operational insights from the Kalamkas field. That work provided an in-depth literature analysis covering the applicability limits for polymer flooding in terms of reservoir temperature, brine salinity, source water selection, oil properties, formation type, and reservoir permeability. The allowable ranges for most parameters are broad and have expanded over the last six decades, reflecting improved technological understanding and field adaptation. However, polymer flooding still faces significant constraints when applied to reservoirs with high temperatures and deep formations.

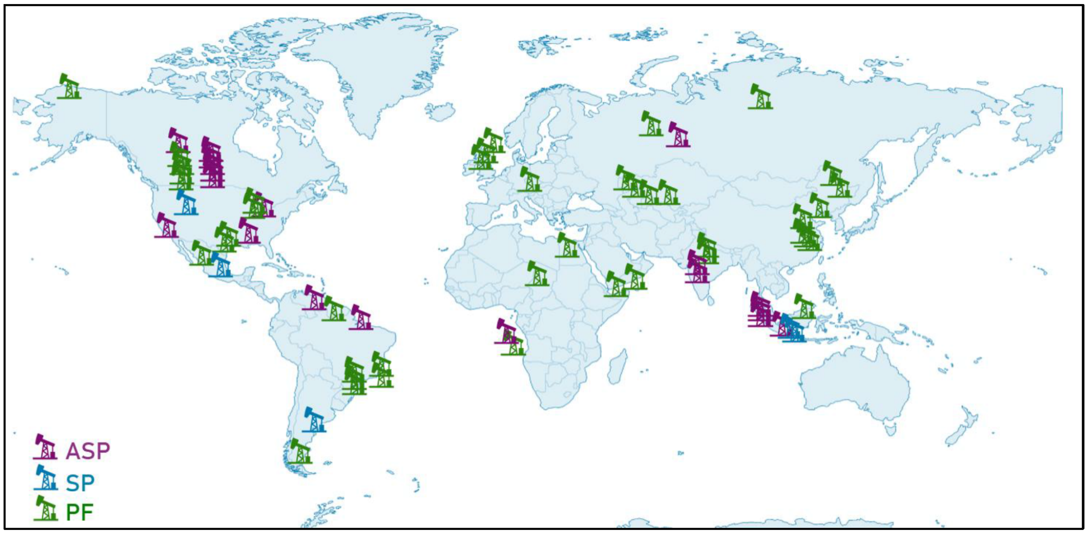

Figure 2—The map of the recent polymer-based enhanced oil recovery projects (P, SP, ASP Flooding) worldwide, updated from (Sagyndikov, Kushekov, et al., 2022)

In this current study, we assess the primary reservoir characteristics of the Uzen field in relation to the established screening criteria for polymer flooding (Fig. 3). As depicted in Fig. 3, Uzen's reservoir properties fall within the acceptable range for all evaluated parameters. Nevertheless, certain conditions, such as elevated formation temperature and high brine salinity, present challenges for polymer selection. Specifically, these harsh conditions necessitate the use of hydrolyzed polyacrylamide (HPAM) polymers that are chemically modified with specific monomers such as AMPS (acrylamido methyl-propane sulfonic acid) or ATBS (acrylamido tertiary-butyl sulfonic acid). Several studies (Diab & Al-Shalabi, 2019; Gaillard et al., 2015; Seright et al., 2021) have demonstrated that incorporating these monomers into the HPAM structure significantly improves polymer resilience to thermal and salinity-induced degradation.

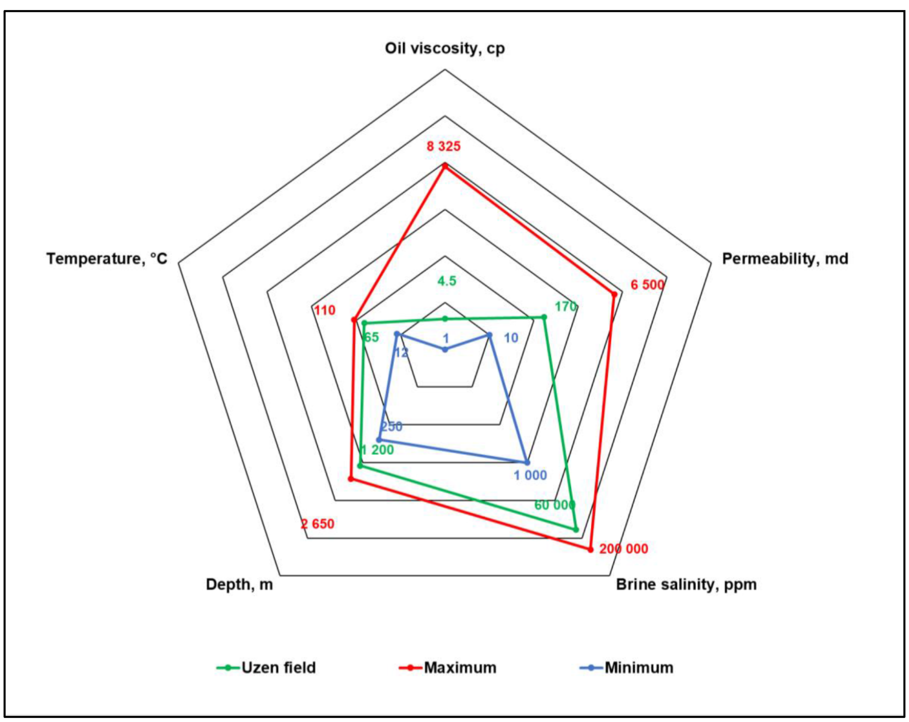

Figure 3—Evaluation of Uzen Field Key Parameters Against Polymer Flooding Suitability Criteria

High reservoir temperatures can accelerate HPAM thermal breakdown, diminishing solution viscosity. Likewise, elevated brine salinity – especially when associated with high concentrations of divalent cations such as calcium (Ca^{2+}) and magnesium (Mg^{2+}) – adversely impacts polymer performance. These cations interact with hydrolyzed polymer chains, causing a reduction in hydrodynamic volume, which in turn decreases the polymer solution's viscosity and may even lead to polymer precipitation (Levitt & Pope, 2008; Moradi-Araghi & Doe, 1987).

Water Source Selection

Caspian seawater was initially selected as the primary injection fluid during the early stages of the Uzen field's development. However, with the field's progressive expansion and the challenges posed by its complex surface topography, which restricted infrastructure placement, a decision was made to incorporate produced formation water into the injection strategy. Consequently, roughly half of the field continues to utilize seawater injection, while the remaining areas rely on reinjection of produced water, with both sources being allocated across different sections of the reservoir.

Our assessment of injection water options encompasses Caspian seawater, produced formation water, and a supplementary source – Cretaceous formation brine – available in limited quantities in central parts of Uzen (Table 1). The Cretaceous brine is characterized by lower salinity levels and a chemical profile resembling that of seawater, making it a potential substitute for produced water in future injection schemes. To complement this compositional comparison, a qualitative risk matrix (Table 5) was developed to assess the three water sources (Table 2).

Table 1—Chemical Characteristics of Water Sources Available at Uzen Field.

Parameters	Unit	Cretaceous brine	Caspian seawater	Produced water
pH		8.4	8.1	6.9
Density	g/cm^3	1.006	1.009	1.041
Ca^{2+}	ppm	217.4	260.8	3 063.7
Mg^{2+}	ppm	104.0	685.1	831.5
$Na^{+}+K^{+}$	ppm	3 488.2	4 134.8	18 856.4
Cl^{-}	ppm	4 230.9	6 325.6	36 660.8
SO_4^{2-}	ppm	2 330.4	3 146.3	30.5
CO_3^{2-}	ppm	36.0	36.0	0
HCO_3^{-}	ppm	122.0	244.0	390.4
Total salinity	ppm	10 528.9	14 832.6	59 883.3
Dissolved oxygen	ppm	0.08	4.92	0.21

Table 2—Risk Matrix of Water Sources

Risk	Cretaceous brine	Caspian seawater	Produced water
Presence of divalent cations (Ca^{2+}, Mg^{2+})	Optimal level	Medium level	High content
Total salinity	Optimal level	Medium level	High content
Dissolved oxygen content	Optimal level	High content	Medium level
TSS and residual oil	Low content	Low content	High content
Treatment cost	Optimal	Medium	High
Water availability	Medium	Optimal	Optimal
Environmental constraints	Low risk	Medium risk	Medium risk

Chemical analysis reveals that while Cretaceous brine has slightly lower salinity compared to seawater, there is a marked difference in total hardness values. Based on these findings, the Cretaceous formation brine is identified as the preferred water source for polymer flooding applications. Should this brine be inaccessible in certain pilot zones, Caspian seawater is considered a secondary option. However, utilizing seawater would necessitate additional treatment processes, either chemical or gas-based, to reduce dissolved oxygen concentrations to negligible levels. The use of produced water is deemed unsuitable due to its high content of total suspended solids (TSS) and residual oil, which compromises its quality for polymer preparation.

In addition to compositional analysis, field investigations were carried out to measure dissolved oxygen levels across all water sources, as oxygen content plays a critical role in maintaining polymer stability. Research by Seright & Skjevrak (2015) highlights that HPAM degradation can be mitigated if dissolved oxygen is kept below 200 ppb, while Jouenne et al. (2017) suggests an even stricter threshold of 46 ppb. Moreover, polymer solutions exposed to air during preparation suffer a substantial loss in viscosity – up to 50% – as evidenced by comparative tests on polymer injection units at Kalamkas, with and without nitrogen blanketing systems (Sagyndikov, Salimgarayev, et al., 2022). These findings underscore the necessity of equipping open polymer maturation tanks with nitrogen blanketing to eliminate air exposure, which could lead to a 25% reduction in polymer powder consumption. It should also be considered that if oxygen is already dissolved in the brine before it reaches the polymer injection unit, nitrogen blanketing alone will not be effective, and additional deoxygenation measures will be required.

Several approaches have been proposed to address the issue with the dissolved oxygen in polymer's brine, including: (1) chemical or mechanical treatment of the process water to remove all iron from the solution (Levitt et al., 2011), (2) the use of chemical additives such as free-radical scavengers or pH adjustment (Gaillard et al., 2010; Wellington, 1983), (3) maintaining dissolved oxygen at an undetectable or acceptable level – as close to zero as possible (Seright et al., 2010), and (4) taking no action (Wang et al., 2008), as demonstrated in the case of the Kalamkas field. The last approach will significantly harm economics due to the loss of polymer viscosity or the need for higher concentrations. In practice, maintaining dissolved oxygen at an acceptable level (below 200 ppb) is considered the most effective and economically viable solution.

Polymer Selection

The experience of the Mangala field in polymer selection has been thoroughly studied, as the field conditions are similar to those of Uzen. Table 3 presents the main complicating factors encountered when using standard hydrolyzed polyacrylamide without ATBS monomers – Flopaam 3630S grade polymer (Shankar et al., 2023). At the Mangala field, a decision was made to change the type of polymer used due to high operational costs. Due to the high reservoir temperature and salinity, the polymer Flopaam 3630S quickly lost viscosity in the reservoir and broke through to the production wells. After conducting laboratory and field studies, the authors concluded that the primary cause of complications was the polymer's low thermal and salt resistance. Since 2020, a decision was made to replace Flopaam 3630S with a more stable polymer containing 5-15% ATBS monomers (FP 5115 VHM, FP 5205 VHM – SNF SA products). Based on the experience in the Mangala field, it was decided to select a polymer with 5% ATBS monomers, Flopaam 5205 VHM, for the Uzen oilfield conditions, due to the reagent's relative cost-effectiveness and the confirmed efficiency of this type in early coreflood studies.

Table 3—Experience of using different polymer types at the Mangala field.

Injection of 3630S polymer (2014-2020)	Injection of polymer with 5-15% ATBS (2020-…)
Polymer solution viscosity at surface – 22-24 cP. Upon breakthrough from production wells – 12-14 cP.	Resistant to hydrolysis at high temperature and salinity
Degree of hydrolysis at surface – 30%. In the reservoir +20-25%, thus when taking reservoir polymer samples, the degree of hydrolysis was 50-55%.	The presence of ATBS monomer groups increases resistance to shear rate
The polymer breakthrough occurred faster (120 days) than predicted during modeling.	Contributes to reduced polymer adsorption
Reduction of oil production rate (OPR) in production wells due to polymer breakthrough.	
Decrease in permeability due to polymer precipitation, resulting in the formation of zones with abnormally high pressure in the reservoir.	
Need to increase polymer concentration, which increases operational costs and injection pressure.	

To better understand the mechanism of polymer filtration in the reservoir, assess risks, and consider polymer properties in hydrodynamic calculations, coreflood studies were conducted to determine the dynamic retention (adsorption & mechanical entrapment) of the Flopaam 5205 VHM polymer versus conventional HPAM without ATBS monomers.

Coreflood studies to determine the level of dynamic adsorption were carried out according to a specially developed methodology:

1. Inject synthetic brine without tracer through 3-5 pore volumes of the core at a rate of 0.5 cm^3/min until stabilized differential pressure values are reached for full saturation of the core with brine;
2. Inject at least 5-13 pore volumes of polymer solution with KI tracer at a concentration of 20 ppm until the polymer concentration Cp at the outlet in the fluid reaches the initial (injected) concentration Co (injection rate 0.5 cm^3/min). The tracer is added to the brine, i.e., before dissolving the dry polymer in it;
3. Effluent samples for concentration measurements are taken every 0.33 PV for the first 5 samples. Then, samples are taken every 1 PV. Simultaneously, two samples of 4-5 cm^3 (each) are collected: first for measuring viscosity and KI tracer concentration on a UV spectrophotometer (wavelength 230-232 nm), and second for measuring polymer concentration on a Shimadzu TOC TNM-L;
4. After determining the concentration values, a C_p/C_0 graph versus PV injection is plotted. The area between the tracer and polymer concentration curves defines the dynamic adsorption (polymer retention). Calculations are performed according to the following equation:

$$R_{pRet} = \frac{\left(\sum\left[\left(C_{poly} * \Delta PV\right) - \left(C_{trac} * \Delta PV\right)\right]\right) + IPV}{m_r} * PV$$

R_{pRet} – dynamic adsorption (retention), µg/g
C_{poly} – polymer concentration in effluent, ppm
C_{trac} – tracer concentration in effluent, ppm
IPV – inaccessible pore volume (assumed as IPV = 0)
PV – pore volume, cm^3
$\Delta\ PV$ – sample volume, cm^3
m_r – core weight, g

The research results are presented in Fig. 4-6. According to calculations using three methods to determine polymer concentration in the effluent, polymer retention for Flopaam 5205 VHM shows the lowest value, indicating potential best performance during a polymer flood. Thus, considering the experience of the Mangala field (an analog field with extensive experience in polymer flood) and the results of these laboratory studies, Flopaam 5205 VHM is selected as a potential polymer for the Uzen field polymer flood pilot.

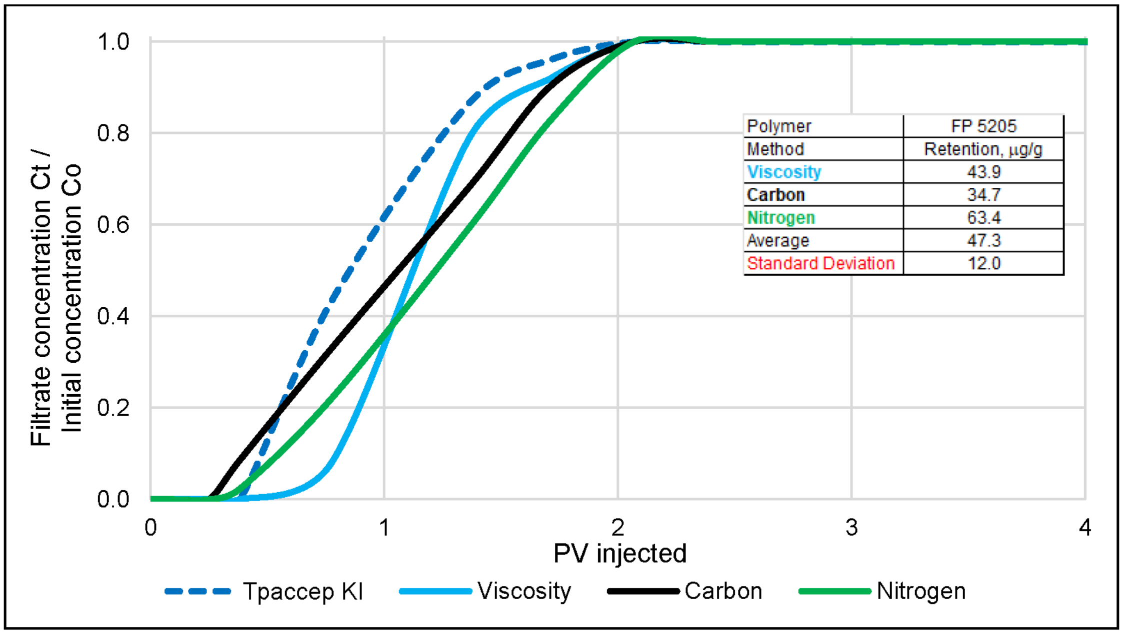

Figure 4—Dynamic retention test of Flopaam 5205 VHM polymer under Uzen field conditions (permeability 100 mD, T = 62°C, polymer concentration 1534 ppm).

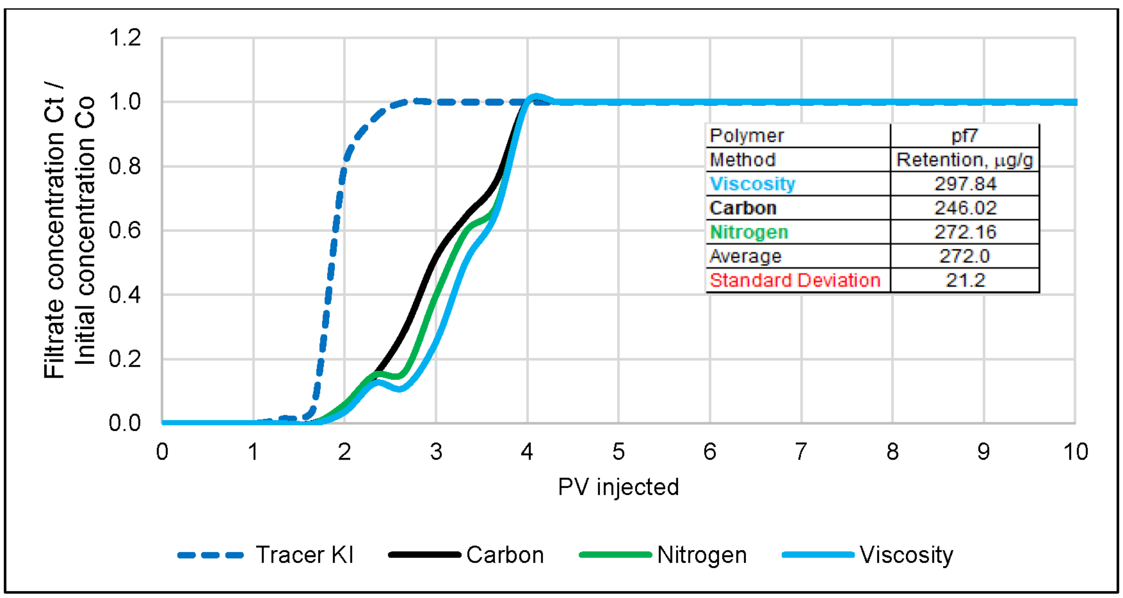

Figure 5—Dynamic retention test of pf-7 polymer (without ATBS) under Uzen field conditions (permeability 100 mD, T = 62°C, polymer concentration 1534 ppm).

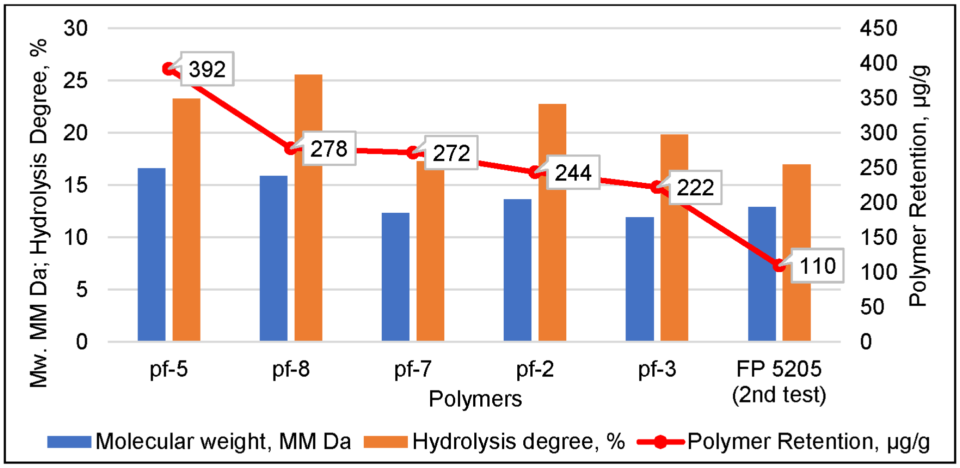

Figure 6—Dynamic retention tests for different polymers including second test of Flopaam 5205 VHM.

Theoretically, less retention is expected as permeability increases. Therefore, experience-supported recommendations for polymer selection depending on polymer weight have been made by Wang et al. (2009). The minimum permeability required for successful polymer flooding is in the range of 100-300 md, and MW should generally be not greater than 17-25 million Daltons. This statement is supported by Sagyndikov & Kushekov et al. (2022) based on actual field applications, where the permeability is mostly greater than 100 md, while the average permeability is around 2 000 md. However, Song et al. (2022) showed promising laboratory results, where HPAM can effectively propagate through the tight low permeable (<50 md) carbonate rocks. The novel polymers can extend the minimum applicability range of permeability, and it has high relevance for future research & development.

Target polymer viscosity determination

A sector from the dynamic simulation model of the Uzen field was utilized for this study. The model, which includes two injectors and ten producers, was history-matched against the latest production data to ensure an accurate representation of current reservoir conditions. The history match achieved a high degree of accuracy, with discrepancies between the model and historical data of just 0.9% for cumulative liquid and 4.2% for cumulative oil (Fig. 7). The optimal polymer viscosity was then determined through a sensitivity analysis using the "elbow" method. To isolate the specific impact of viscosity, all other reservoir and operational parameters were held constant while a range of polymer viscosities (3 to 40 cP) was evaluated. Thus, reservoir properties such as well interference and heterogeneity remained unchanged from the base waterflooding scenario.

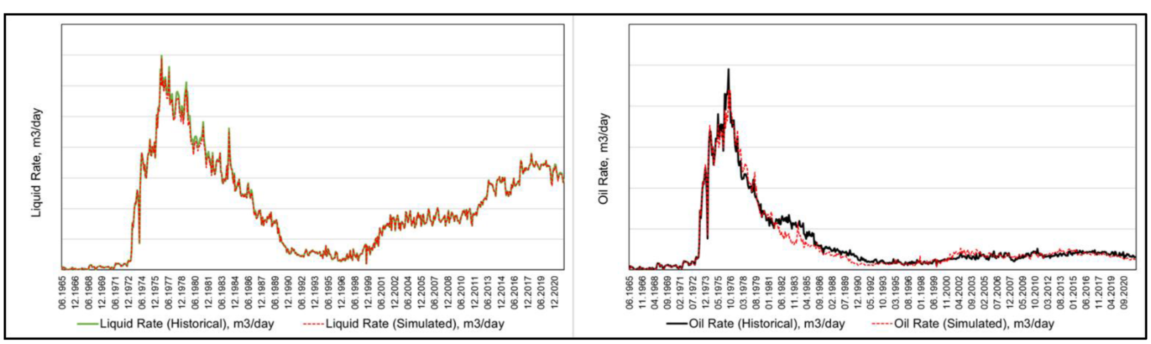

Figure 7—History matching plots for Uzen field

By plotting the resulting rate of production increase against each corresponding viscosity value, the point of diminishing returns was identified, defining the optimal target for injection. The rationale for utilizing the rate of production increase, rather that the cumulative incremental oil volume from each case, stems from the inherent uncertainty associated with absolute recovery values predicted by a model. This approach allows for a more efficient determination of the optimal viscosity by focusing on the immediate system response. It thereby mitigates the complexities and compounding uncertainties of long-term volumetric forecasting. Consequently, relying solely on the model's absolute predictions for total incremental oil would introduce an unacceptable level of project risk.

It is clearly shown on Fig. 8 that production increases sharply as the polymer viscosity rises from 3 to 15 cP. Beyond the 15 cP, the curve flattens considerably, indicating that further increases in viscosity yield only marginal improvements in the production rate. This inflection point, or "elbow', suggests that 15 cP is the optimal target in-situ viscosity. It represents the most efficient choice, providing the maximum production benefit before the economic and operational cost of higher viscosity outweigh the minimal additional gains.

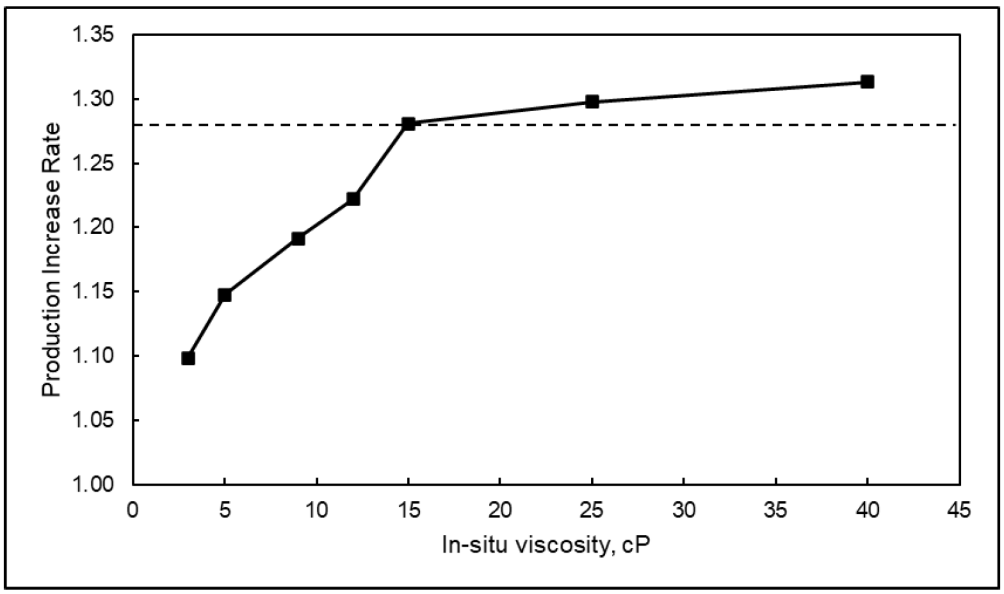

Figure 8—Determination of optimal polymer viscosity.

In cases where a numerical model is unavailable or impractical to use, the target viscosity can be determined using the base-case method developed by Seright (2017). This method is an optimal analytical approach that incorporates key geological and fractional flow parameters. Specifically, it considers geological factors such as permeability contrast alongside fractional flow properties like relative permeability and fluid viscosities, which together determine the mobility ratio. While this analytical method provides a solid foundation, a numerical simulation model is not limited to a few parameters, as it holistically incorporates a vast of range of dynamic data including detailed reservoir geology, PVT properties, field development history, well interference patterns, and the properties of polymer. Therefore, determining the optimal viscosity using a comprehensive hydrodynamic model represents a more objective and reliable method.

With the target viscosity determined, the subsequent step is to establish the polymer concentration required to achieve this value considering operational water for solution. This is accomplished by conducting rheological studies on the selected polymer Flopaam 5205. As illustrated in Fig. 9, a polymer solution with an active concentration of 1426 ppm exhibits a viscosity slightly exceeding the 15 cP target within the shear rate range corresponding to reservoir velocity. This technically validated concentration will be adjusted to a practical value of 1600 mg/L for field implementation to streamline mass calculations and simplify operations.

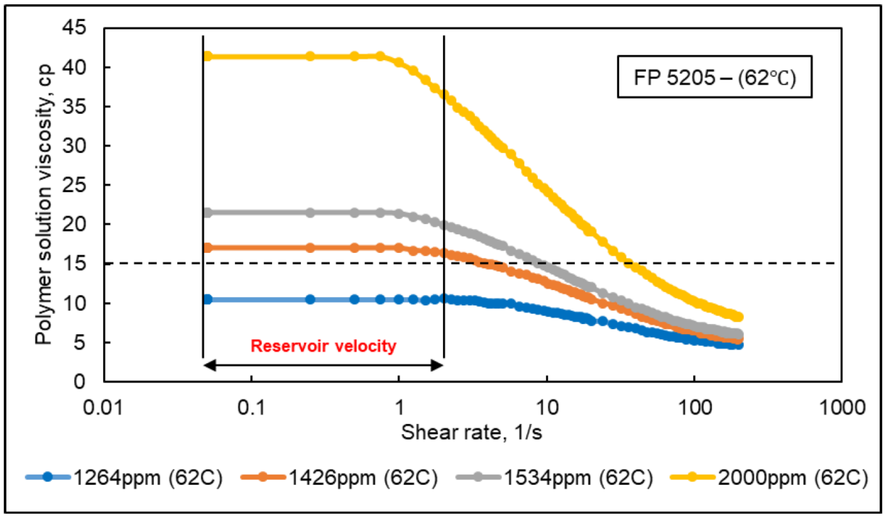

Figure 9—Rheological results of Flopaam 5205 at reservoir temperature.

To facilitate daily quality control (QC) of the polymer solution in the field, rheological studies were also performed under surface conditions (25 °C and a shear rate, γ, of 7.34 s^{-1}). As shown in Fig. 10, the solution exhibits a viscosity of approximately 25 cP under these conditions. Therefore, this value is established as the field-b ased QC target for the operations team to ensure the injected fluid consistently meets specifications.

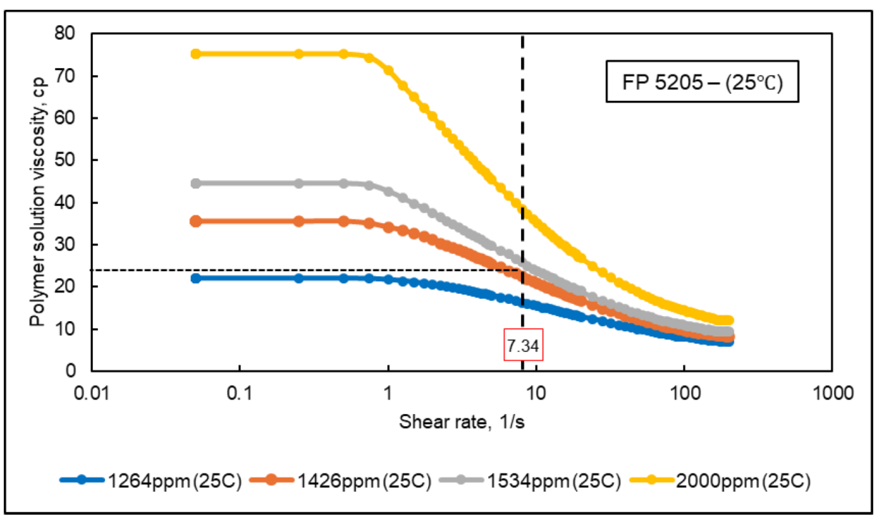

Figure 10—Rheological results of Flopaam 5205 at surface temperature.

The polymer flooding suitable spots and pilot selection

The geological conditions for the application of polymer flooding generally coincide with those for conventional waterflooding. However, there are specific differences. The method may prove economically inefficient in oil reservoirs underlain by strong aquifers, or those with extensive oil–water or gas–oil contact zones, as well as in fields with conductive tectonic faults. At the same time, polymer flooding can be an effective method under conditions where conventional waterflooding is unprofitable, such as in cases of high-viscosity oils or reservoirs with anomalously high heterogeneity (Ning et al., 2020; Zhao et al., 2020).

Table 4 presents the screening results for the upper and lower horizons of the Uzen field and its satellite analogue, the Karamandybas field. The screening results indicate that horizons XIII–XVIII offer the most favorable conditions for polymer flooding. At the Karamandybas field, as well as in horizons XIX–XXIV of the Uzen field, injectivity, permeability, reservoir temperature, and formation water salinity do not fully meet the preferred criteria range. However, noncompliance with the screening criteria does not necessarily preclude the application of polymer flooding, as the deviations from the optimal values are relatively minor. It should be noted, nevertheless, that polymer flooding in these reservoirs is unlikely to achieve the same effectiveness as in horizons XIII–XVIII. Furthermore, the data clearly indicate that reservoir temperature increases with depth. Consequently, the most favorable horizons for pilot testing are the shallower XIII–XVIII horizons, which have the lowest temperatures, thereby reducing the thermal impact on HPAM within the reservoir.

Table 4—Geological and petrophysical screening of reservoirs in the Uzen and Karamandybas fields

Parameters	Unit	Criteria					
		Min.	Max.	Optimum range	Uzen, XIII-XVIII Horiz.	Uzen, XIX-XXIV Horiz.	Karamandybas
Lithology	type	sandstone	carbonate	sandstone	sandstone	sandstone	sandstone
Water injectivity	m³/d	100	unlim.	>200	347	164	156
Permeability	mD	10	unlim.	>200	210	60	180
Porosity	%	10	37	>20	24	20	19
Net pay thickness	m	3	50	>10	25	46	13,5
Temperature	°C	40	100	<80	60	80	72
Reservoir depth	m	150	3000	<2750	1116	1410	1640
Oil viscosity	cP	1	2675	<1000	4,07	4,24	4,1
Formation water salinity	g/L	0,4	167	37	50,5	64	64
Mobility ratio	-	1	40	<10	1,7-2,7	1,7-2,7	1,7-2,7
Oil saturation	%	50	92	>40	57	60	61
Water cut	%	unlim.	96	<95	89,8	77	86,2

(dark green)	Meets preferred criteria
(light green)	Partially meets preferred criteria
(yellow)	Does not meet preferred criteria

The target zones of perspective shallower XIII–XVIII horizons are predominantly represented by fluvial systems – sandstone-siltstone lenticular bodies deposited during the Miocene-Pliocene under fluvial sedimentary conditions. These reservoirs, encased in clay-rich formations, exhibit significant heterogeneity in thickness, composition, and permeability (up to several hundred mD).

Key characteristics of fluvial sandstones include:

- high permeability and porosity, providing favorable conditions for hydrocarbon migration and accumulation;
- lenticular geometry with interbedded clay layers, creating isolated oil-saturated zones;
- variability in thickness, permeability and oil saturation.

Fluvial deposits are the primary oil-bearing reservoirs in the Uzen field; however, their heterogeneity often results in non-uniform fluid distribution during waterflooding. Early water breakthrough through high-permeability zones leaves significant residual oil in low-permeability intervals. These conditions necessitate the use of selective EOR technologies, such as polymer flooding, to enhance sweep efficiency and maximize ultimate oil recovery.

According to Seright (2017), polymer flooding has demonstrated higher effectiveness in reservoirs that are monolithic with the vertical cross-flow and exhibit good lateral continuity compared to compartmentalized or highly segmented formations without vertical cross-flow. For the fluvial systems of the Uzen field, this is particularly relevant: when selecting areas for polymer flooding, priority is given to zones with more continuous and well-connected reservoir bodies, which supports improved areal sweep and oil recovery.

Candidate area screening

A detailed facies analysis of the target horizons was conducted to identify the optimal candidate for a polymer flooding pilot test (Table 5 — Candidate area screening in the Uzen Field). A total of 22 potential areas were evaluated, with the majority (15 out of 22) located in Horizon XIV, as shown in Fig. 11. The screening results indicate a high concentration of suitable fluvial systems within this horizon, making it the primary candidate for field-scale pilot implementation.

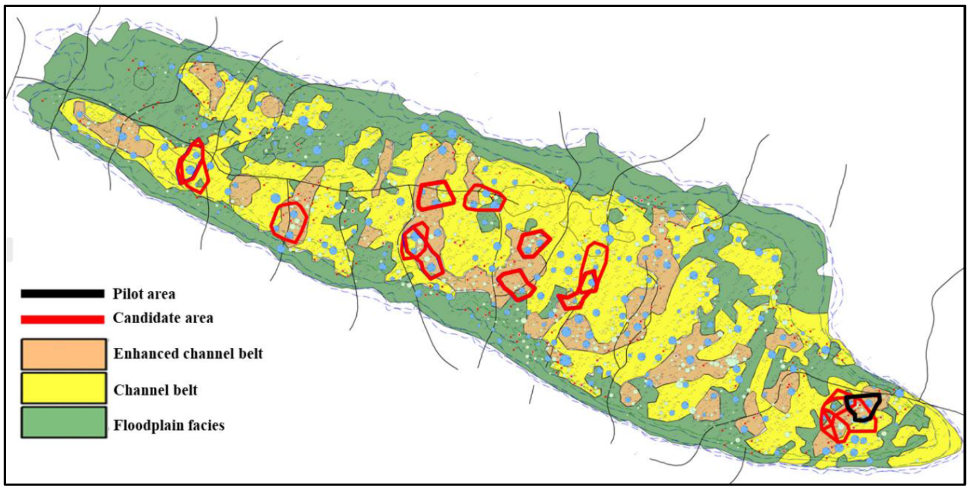

Figure 11—Geological map of Horizon XIV, Uzen field, indicating potential areas for polymer flooding.

Table 5—Candidate area screening in the Uzen Field

№	Area	Number of injectors, unit	Number of producers, unit	WCT, %	RF, %	Well pattern	Facies	Availability of Cretaceous brine
1	14_1	3	22	93.8	80.5	Requires infill drilling of 2–3 wells	Channel belt	Unavailable
2	14_2	2	15	88.9	77.7	Requires infill drilling of 2–3 wells	Predominantly channel belt	Available
3	14_3	2	15	92.3	84.8	Well pattern established	Predominantly channel belt	Unavailable
4	14_4	3	15	91.9	80.9	Requires infill drilling of 2–3 wells	Predominantly channel belt	Unavailable
5	14_5	3	20	90.1	80.2	Requires infill drilling of 4–5 wells	Channel belt	Unavailable
6	14_6	3	18	90.6	72.4	Requires infill drilling	Channel belt	Available
7	14_7	4	30	83.4	75.5	Requires infill drilling	Enhanced channel belt	Available
8	14_8	2	19	92.5	83.0	Requires infill drilling of 2–3 wells	Floodplain facies	Available
9	**14_9**	**3**	**24**	**91.2**	**71.7**	**Established**	**Enhanced channel belt**	**Available**
10	13_1	2	12	92.4	68.0	Established	Channel belt	Unavailable
11	13_2	4	22	89.9	56.8	Requires infill drilling	Channel belt	Unavailable
12	17_1	3	14	87.6	75.2	Requires infill drilling	-	Unavailable
13	14B-1	2	17	91.2	74.5	Significant interwell spacing	Channel belt	Unavailable
14	14B-2	3	20	94.3	81.4	Established	Channel belt	Unavailable
15	14B-3	3	18	92.4	71.5	Requires infill drilling	Channel belt	Unavailable
16	14B-4	2	15	91.3	71.5	Established	Channel belt	Available
17	14B-5	2	14	93.1	80.3	Requires infill drilling of 2–3 wells	Channel belt	Unavailable
18	14B-6	2	17	92.4	73.0	Established	Channel belt	Unavailable
19	17B-1	2	22	92.1	66.6	Established	-	Unavailable
20	17B-2	2	13	90.5	63.5	Established	-	Unavailable
21	16B-1	2	12	87.9	77.3	Requires infill drilling	-	Unavailable
22	15B-1	2	8	94.7	74.6	Requires infill drilling	-	Unavailable

As a result of a comprehensive evaluation of geological and petrophysical characteristics, facies architecture, and reservoir dynamics, Area 14_9 has been identified as the most promising candidate for a polymer flooding pilot. This selection is based on a combination of favorable reservoir properties, the availability of Cretaceous brine, and compliance with key development criteria, all of which together provide strong potential for achieving high sweep efficiency and incremental oil recovery.

In addition, Areas 13_1 and 14B-6 have been highlighted due to their established well patterns and location within the channel belt of the field, while currently exhibiting low oil production rates. These areas may be considered as follow-up candidates should the pilot at Area 14_9 prove successful. However, the absence of the Cretaceous brine water source at these sites will require the development of appropriate technical solutions.

The pilot area comprises of 3 injection wells planned to switch from waterflood to polymer flood: PF-1, PF-2, and PF-3 and 24 offset production wells (P-1…P-24). Offset producers confirmed through the streamline simulations shown in Fig. 12.

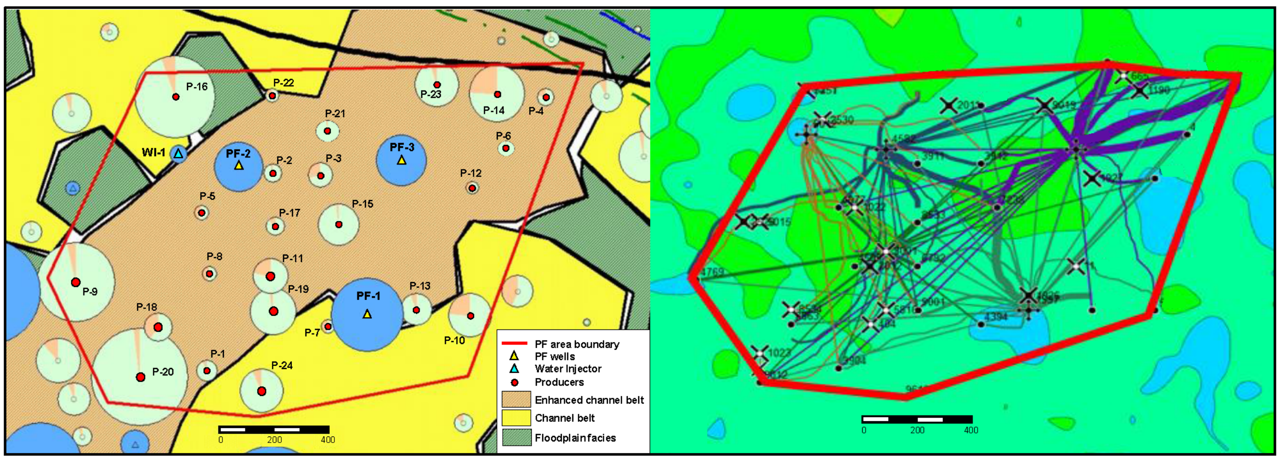

Figure 12—Polymer flood pilot area bubble map and streamline simulation for selecting offset producers

Correlation analysis of the area identified a laterally continuous sand body within Unit 14-B (Fig. 13). Both production and injection wells penetrate the low-permeability Unit 14-A (~20 mD) and the top of the sand bodies of Units 14-B and 14-C, which contain medium- to high-permeability reservoirs (~213 mD). The majority of well perforations are completed within Unit 14-B. These sand bodies range from 5 to 40 m in thickness, with the largest portion of recoverable reserves concentrated in the paleo-channel Unit 14-B. The paleo-channel zone exhibits greater lateral continuity and reservoir homogeneity, whereas the inter-channel area consists predominantly of low-permeability interbeds. This geological configuration aligns with the findings of Seright (2017), which indicate that polymer flooding is most effective in monolithic, well-connected reservoirs, thereby supporting the selection of Pilot Area 14_9 as the target for the polymer flooding pilot project.

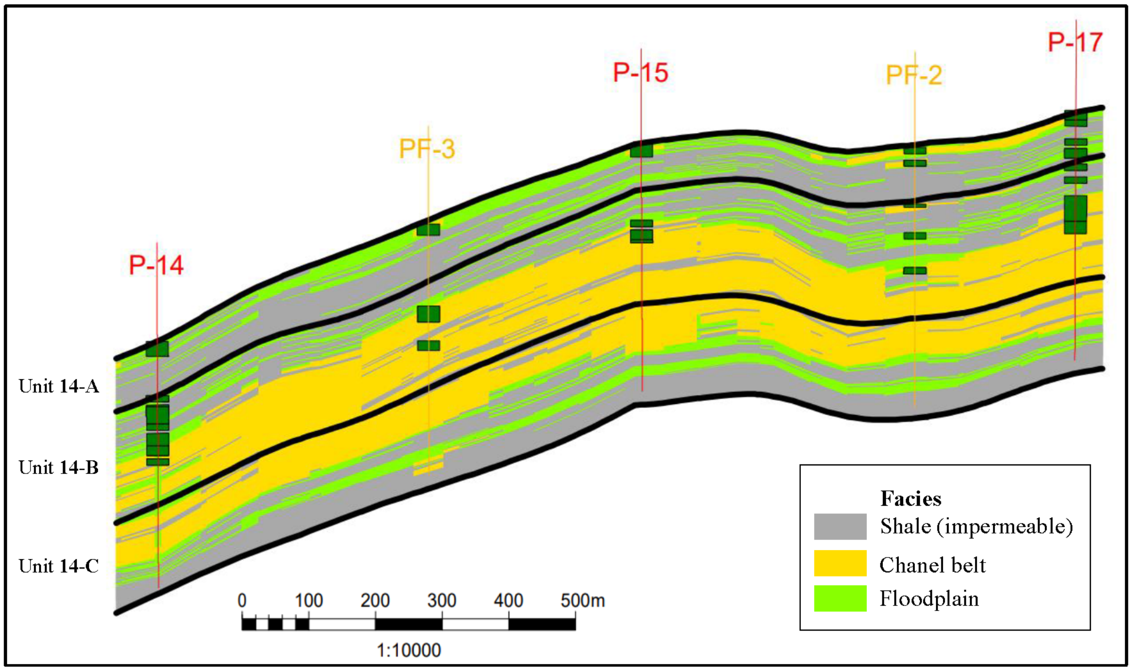

Figure 13—Geological cross-section of the PF pilot area.

Feasibility and field performance: why polymer flooding outperforms ASP

ASP project economics

Prior to discussing the technical challenges of ASP flooding, it is crucial to address its significant cost disadvantage compared to standard polymer flooding. The cost breakdown of the chemical components illustrates this disparity. On the industrial market, HPAM polymers are priced at approximately $2,000- $ 3,500 per ton, while surfactants are typically twice as expensive, at around $7,000 per ton. The required alkaline agent contributes an additional $650 per ton to the formulation. Consequently, the large volumes needed for this three-component system can lead to substantial expenditures on chemicals alone, a factor highlighted by the high cost observed in pilot-scale ASP projects (Sagyndikov, Kushekov, et al., 2022).

To support this claim, we reference an economic analysis of the Daqing field ASP project (Gao et al., 2020), which was detailed in our previous work (Sagyndikov, Kushekov, et al., 2022). The findings reveal that the project's operational expenditures far exceed the revenue generated from incremental oil recovery. Specifically, the authors state that the economic benefit of the project was $32.35 million (at an oil price of $50/bbl), which is approximately $10 million less than the chemical costs alone (Table 6, Table 7). It is critical to note that this calculation omits all other expenses; the total cost would be significantly higher when factoring in CAPEX and other operational costs.

Table 6—Averaged design of Daqing (China) ASP flooding pilot according to Guo et al. (2017) and Gao et al. (2020).

1st year		2nd -4th years		5th year		6th year		
Pre-Slug (polymer)		ASP Main Slug		ASP Vice Slug		Post-Slug (polymer)		Total injected
PV	Concentration, %	PV	Concentration, %	PV	Concentration, %	PV	Concentration, %	PV
0,2	0.14	0.505	0.3%S+1%A+0.18%P	0.21	0.1%S+1.2%A+0.16%P	0.18	0.12	1.0924

Table 7—Cost breakdown of chemicals in Daqing ASP project based on average market costs.

Slug consequence	Chemicals	Injected weight, tons	Cost for chemicals, USD	Cost for chemicals over the pilot period, USD
Pre-Slug (polymer)	A	0	0	1 762 236
	S	0	0	
	P	503.50	1 762 236	
ASP Main Slug	A	9 080.91	5 902 592	30 693 476
	S	2 724.27	19 069 911	
	P	1 634.56	5 720 973	
ASP Vice Slug	A	4 479.68	2 911 789	7 615 449
	S	373.31	2 613 144	
	P	597.29	2 090 515	
Post-Slug (polymer)	A	0	0	1 357 929
	S	0	0	
	P	387.98	1 357 929	
Total				41 429 089

Based on this evidence, it can be concluded that such projects are often undertaken for academic and research purposes rather than commercial benefit. The primary goal is to gain a better understanding of the complex mechanisms involved, a view supported by Dean et al. (2020).

ASP field performance

Another significant source of cost overruns during ASP field implementation stems from severe operational challenges. These primarily include the scaling of both subsurface and surface equipment (tubing, artificial lift systems, heaters etc.), and the formation of stable, hard-to-break emulsions at production facilities. Many authors revealed often artificial lift system failures during ASP because of scaling issues (Finol et al., 2012; Jain et al., 2020; Volokitin et al., 2017) (Fig. 14). The primary cause of scale formation is the significantly high pH level, caused by the large amounts of injected alkali. Mitigating this problem necessitates costly interventions. These include redesigning production facilities (Gang et al., 2007), changing the artificial lift method, as it was done for Mangala ASP project (Pandey et al., 2016), or implementing aggressive and expensive scale inhibitors (Cheng et al., 2014).

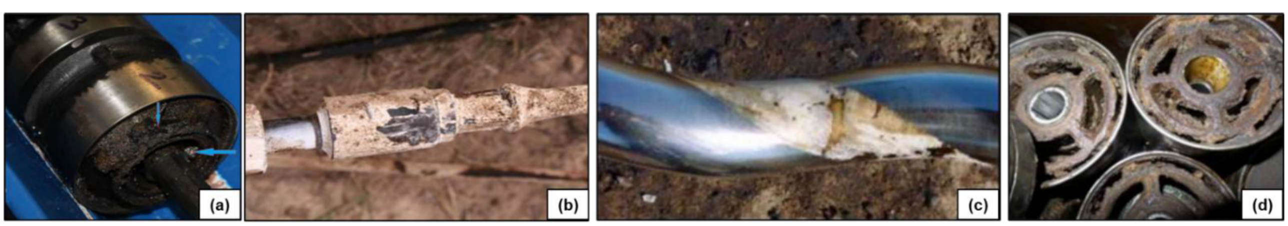

Figure 14—Emulsion and scalings during ASP flooding: (a) emulsion in ESP (Pandey et al., 2016) (b) scaling on rod pump (Jiecheng et al., 2008) (c) scaling on PCP rotor (Guo et al., 2017) (d) scaling deposits on ESP wheels (Volokitin et al., 2017)

The formation of viscous, hard-to-break emulsions is another operational challenge, as observed in several Chinese pilot projects. The severity of this issue is highlighted by reports of produced fluid viscosities reaching as high as 487 cP – in some cases, exceeding the injected ASP solution viscosity by a factor of ten (Guo et al., 2017). While the phenomenon is not yet fully understood, its primary consequence is a tangible loss of production. Therefore, the potential emulsification issues must be proactively addressed during the project design phase. Best practices include both field-level strategies, such as installing dedicated demulsifier injection wells near producers (Jain et al., 2020), and preliminary lab work to identify cost-effective demulsifiers (Finol et al., 2012).

An underlying factor contributing to these operational challenges is the crude oil's Total Active Number (TAN). This property, measured by the amount of potassium hydroxide (KOH) needed to neutralize the acids in the oil (in mg KOH/g), governs the crude's natural tendency to form emulsions when reacting with chemical agents. A sufficiently high TAN (>0.20 mg KOH/g according to Guo et al. (2017)) is generally considered beneficial, as it can react with the injected alkali to create in-situ soap, enhancing the overall effectiveness of the surfactant and leading to better oil mobilization. Table 8 provides a summary of TAN values from various ASP projects. As the data indicates, even in projects with an adequately high TAN, severe operational complications can still arise. This suggests that while a favorable TAN is advantageous, it does not eliminate the inherent risks associated with ASP flooding.

Table 8—Overview of oil TAN at ASP projects

Oilfields	Oil TAN, mg KOH/g	ASP flood conducted	Incremental RF, %	Complications
Bhagyam, India	2.00	Yes	20	Emulsion, scaling, corrosion
Al Khalata, Oman	0.78	Yes	-	Emulsion, scaling
Karazhanbas, Kazakhstan	0.251	No	-	-
Kalamkas, Kazakhstan	0.132	No	-	-
Uzen, Kazakhstan	**0.048**	No	-	-
West Salym, Russia	0.040	Yes	16	Scaling
Daqing, China	0.020	Yes	>20	Emulsion, scaling, repairment of surface equipment

The feasibility of the polymer flooding pilot

The incremental oil production from polymer flooding is calculated as the difference between the actual production under polymer flooding and the baseline oil production during waterflood. The baseline oil production during waterflood is forecasted based on the displacement characteristic (DC) or decline curve analysis (DCA). The term DC refers to various models for predicting the filtration process during waterflooding of an oil reservoir (or field). Some DC models are based on the Buckley–Leverett theory (Sipachev N.V. & Posevich A.G., 1981), while most have been derived empirically from actual oil field development data. Based on the calculation results, the annual decline rate for the pilot test area was 6.8% (Fig. 15), compared to more than 10% for the field as a whole. Therefore, the calculations are considered reliable for further evaluation of the technology's effectiveness.

The forecast for incremental oil (EOR) production is based on the water cut dynamics as a function of pore volume injected. This relationship is derived from the experience of the Daqing field, owing to its similar geological and physical characteristics to the Uzen field. The core principle of this approach involves forecasting the volume of injected water at the established injectivity of the injection wells. By determining the pore volume throughput from the cumulative volume of injected fluid (water), the water cut behavior during polymer flooding is assessed. The operational factors for injection and production wells are assumed to be identical to those used in the baseline performance forecast.

The project demonstrates strong economic feasibility, achieving payback in the third year of implementation with a profitability index (PI) of 1.93, and a moderate polymer utility factor (50.6 t of incremental oil per 1 t of dry polymer injected), indicating robust returns, making the pilot polymer flooding implementation economically attractive.

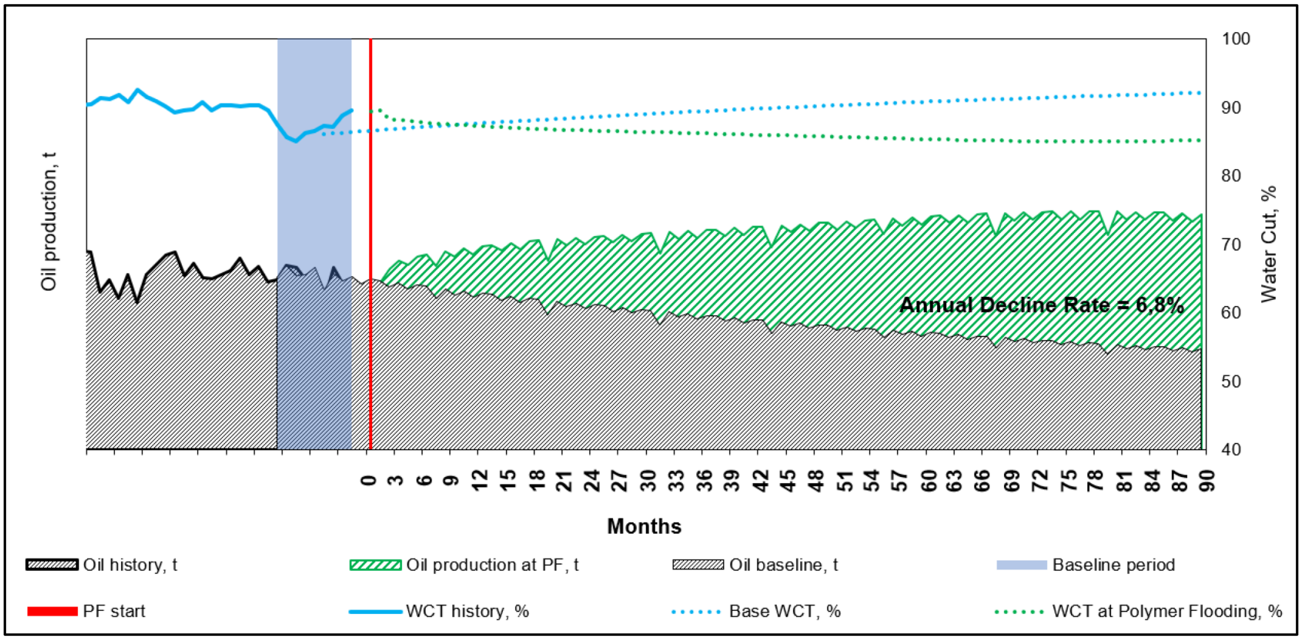

Figure 15—The polymer flood pilot production forecast.

The injection strategy and the pilot surveillance plan

Injection program

The polymer injection program is based on the optimal in-situ viscosity target of 15 cP, as determined in the *Target polymer viscosity determination* section. Rheological studies for Flopaam 5205 VHM under reservoir conditions established that this viscosity corresponds to a polymer concentration of 1,426 ppm. For operational simplicity and to facilitate field mass balance calculations, this value was rounded up to 1,600 mg/L for pilot implementation.

The program considers key process parameters such as total injected polymer solution volume (m^3) and total dry polymer consumption (kg), with a monthly distribution over the analyzed period. It is developed annually during the entire pilot testing period by selecting optimal injection parameters, taking into account the technical capabilities of the polymer injection unit (PIU) and the inventory of injection wells, as well as the possibility to adjust injection parameters – including the viscosity and concentration of the injected polymer solution – if required.

The primary objective of the injection program is to define optimal parameters to maximize efficiency, schedule shutdowns for field investigations, and account for a commissioning/start-up period at the beginning of the pilot (up to three months) to evaluate and fine-tune PIU operation before reaching the planned regime. Another key goal is to establish the daily polymer consumption to enable accurate demand planning, including calculating the required quantities in 25 kg bags. The injection program is planned and adjusted according to the exact daily bag consumption. Specifically, during days 1–10 of the first month, daily consumption will be 22 bags; days 11–20 – 27 bags; days 21–31 – 32 bags; and in after the second month – 48 bags. The injection schedule for the first year is presented in Table 9.

Table 9—The polymer flood injection program

Day/ Month	*Polymer concentration, mg/L	**Viscosity, cP	Injectivity, m^3/day			Dry polymer consumption, kg			Total injectivity, m^3/day	Total dry polymer consumption, kg/month (***bag/day)	Cumulative dry polymer consumption, kg	Injection wells uptime
			PF-1	PF-2	PF-3	PF-1	PF-2	PF-3				
1st-10th day/ 1st	1 000	8-12	200	300	300	1 500	2 000	2 000	800	5 500 (22)	5 500	0,70
11th-20th day/ 1st	1 200	14-18	200	300	300	1 750	2 500	2 500	800	6 750 (27)	12 250	0,70
21st-31st day/ 1st	1 400	15-20	200	300	300	2 000	3 000	3 000	800	8 000 (32)	20 250	0,70
2^{nd} – 8^{th}	1 600	20-25	200	300	300	9 000	13 500	13 500	800	36 000 (48)	56 250	0,95

**Polymer concentration can vary to Δ10% considering the planned injection schedule and the actual downtime of the PIU*
***At a shear rate of 7.34 s^{-1} and T = 25 °C*
**** 1 bag = 25 kg*

Pilot Surveillance

A comprehensive surveillance program is planned to ensure effective monitoring, quality assurance, and performance evaluation of the polymer flooding pilot. The program will cover four key areas (Fig. 16):

1. **Well Surveys**
   Baseline and periodic well surveys will be conducted to assess technical conditions, analyze well logs, and measure temperature profiles. Three-phase flowmeter data and water cut (WCT) measurements will be collected to track production performance and monitor changes in fluid composition over time.
2. **Polymer QA/QC**
   Strict quality assurance and quality control procedures will be implemented throughout the polymer handling and injection process. These will include verification of chemical specifications, inspection of the PIU condition, monitoring polymer dissolution quality, measuring dissolved oxygen levels, and conducting viscosity checks. Viscosity meters will be calibrated regularly, and daily reports will be compiled to document operational status and any deviations.
3. **Pilot Wells Monitoring**
   Continuous monitoring of pilot injection and production wells will be carried out. Planned activities include chemical analysis of produced water, tracking polymer breakthrough and concentration in produced fluids, and using tracers to map fluid movement. Pressure transient analysis (PTA) and downhole fluid sampling (Sagyndikov, Seright, et al., 2022) will be performed to evaluate reservoir response and identify potential conformance issues.
4. **Performance Evaluation**
   The effectiveness of the polymer flooding process will be assessed considering all well events and the base case waterflood performance. Key performance indicators will include enhanced oil recovery factor, utilization factor, changes in recovery factor (ΔRF), and WCT trends. Comparative analysis of diagnostic plots for waterflood and polymer flood responses will provide insights into injectivity, mobility control, and sweep efficiency improvements.

   This integrated surveillance approach is designed to enable proactive decision-making, facilitate timely operational adjustments, and provide a reliable evaluation of the pilot's technical and economic performance. The resulting data set will serve as a strong foundation for potential polymer EOR expansion and full-field implementation.

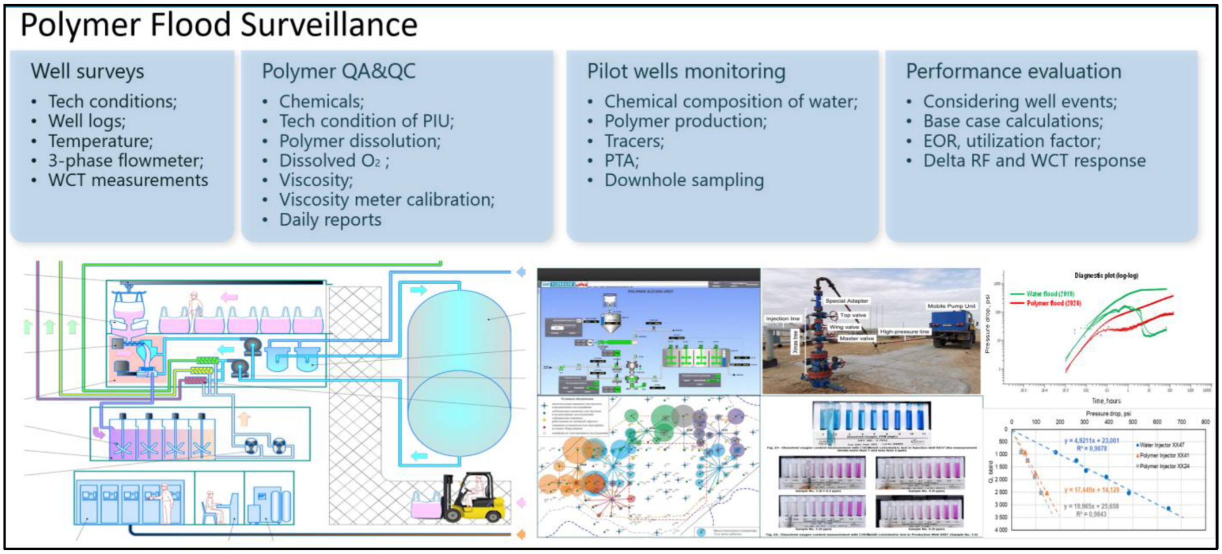

Figure 16—The polymer flood pilot surveillance concept.

Results Discussion

This study highlights how an accelerated laboratory-to-field workflow can overcome traditional bottlenecks in polymer flooding design. By streamlining polymer screening, focusing on the most representative reservoir samples, and integrating rheology with simulation, the approach enabled rapid definition of operational parameters suitable for Uzen's challenging conditions. Importantly, this case illustrates how practical engineering decisions—such as selecting ATBS-modified HPAM and defining a viscosity target through sensitivity analysis—can provide clarity and confidence in project planning even under uncertainty.

Nevertheless, several unresolved aspects remain. Field-scale heterogeneity, particularly within fluvial systems of Horizon XIV, poses risks of uneven sweep and early breakthrough, which may not be fully captured in laboratory or model studies. The reliance on Cretaceous brine as the preferred injection water, while technically advantageous, introduces logistical constraints due to limited availability across the wider field. In addition, polymer stability under prolonged exposure to high salinity and temperature is a potential risk that can only be addressed through extended surveillance.

The broader implication is that polymer flooding in Kazakhstan, a concept long discussed but rarely implemented, can be advanced through workflows that emphasize practicality, cost discipline, and rapid iteration rather than exhaustive laboratory campaigns. Comparisons with international experiences (e.g., Mangala and Daqing) indicate that adapting polymer chemistry to local brine and temperature conditions is often more decisive for success than large-scale ASP-type experiments.

Future work should explore four directions: (1) long-term pilot surveillance to evaluate EOR performance, polymer retention, and reservoir propagation behavior under real reservoir dynamics, (2) strategies for scaling up polymer flood while diversifying water sources to mitigate supply risk, (3) examine the periodical implementation of conformance control (gel treatment) during a polymer flood. and (4) investigation of next-generation polymers designed for higher temperature and salinity to expand applicability to deeper horizons and to achieve less polymer dosage and retention, which can improve project economics. Addressing these points will strengthen the case for transitioning from pilot to full-field deployment and contribute to the broader adoption of polymer EOR in Kazakhstan and Central Asia.

Conclusions

The study presents a comprehensive laboratory-to-field workflow for the accelerated implementation of polymer flooding at the Uzen field, encompassing screening and polymer selection, economic evaluation, and surveillance planning. This integrated approach enabled the rapid transition from initial concept to pilot execution, with key technical and economic milestones achieved within a compressed timeline. The aforementioned comprehensive work is succinctly integrated into a document titled "Project Passport," which can serve as a guide for implementing polymer flooding in other oil fields. The introduction of polymer flooding to the Uzen oil field marks a significant advancement in Kazakhstan's EOR practices. This study presents an innovative framework, distinct from conventional approaches, that offers expedited strategies for future implementation. Recommendations emphasize the integration of novel methodologies and engineering workflows to enhance efficiency, cost-effectiveness, and scalability. The main outcomes of this study are as follows:

1. **Screening confirmed feasibility** – The geological and petrophysical characteristics of the Uzen field, particularly in the shallower XIII–XVIII horizons, fall within the acceptable range for polymer flooding, with favorable injectivity, permeability, and oil saturation.
2. **Polymer selection optimized for harsh conditions** – Laboratory studies and analogue field experience (Mangala) led to the selection of Flopaam 5205 VHM (5% ATBS) as the most suitable polymer, offering improved thermal and salinity resistance compared to standard HPAM polymers.
3. **Optimal operational parameters established** – Numerical simulation using the "elbow method" determined a target in-situ viscosity of 15 cP, achieved with a field concentration of ~1,600 ppm. This provides the best balance between incremental production gains and operational cost.
4. **Water source strategy defined** – Cretaceous formation brine was identified as the preferred injection water due to its lower salinity, hardness, and absence of dissolved oxygen. Where unavailable, Caspian seawater can be used with additional deoxygenation measures to ensure polymer stability. Maintaining dissolved oxygen in injection water at an acceptable level – as close to zero as possible – remains the most effective and economically viable strategy to preserve polymer stability, minimize polymer consumption, and reduce additional OPEX for water treatment.
5. **Economic assessment supports pilot deployment** – The pilot is projected to achieve payback in the third year of operation, with a profitability index (PI) of 1.93 and a polymer utility factor of 50.6 t of incremental oil per ton of polymer injected, indicating robust economic viability.
6. **Comprehensive surveillance program developed** – The plan integrates well surveys, polymer QA/QC, reservoir monitoring, and performance evaluation to enable proactive operational adjustments and reliable assessment of technical and economic performance.

Nomenclature

Symbol/Abbreviation	Unit	Quantity
bbl	Barrel	Volume
°C	Degrees Celsius	Temperature
cm3	Cubic centimeter	Volume
cP	Centipoise	Viscosity
g	Gram	Mass

Symbol/Abbreviation	Unit	Quantity
g/cm3	Grams per cubic centimeter	Density
g/L	Grams per liter	Concentration/Salinity
kg	Kilogram	Mass
m	Meter	Length/Depth/Thickness
m3	Cubic meter	Volume
m3/d or m3/day	Cubic meters per day	Flow Rate/Injectivity
mD	Millidarcy	Permeability
mg KOH/g	Milligrams of KOH per gram	Total Active Number (TAN)
mg/L	Milligrams per liter	Concentration
MMBBL	Million barrels	Volume
ppb	Parts per billion	Concentration
ppm	Parts per million	Concentration
s^{-1}	Inverse seconds	Shear Rate
t	Tonne (metric ton)	Mass
USD	United States Dollar	Currency
wt.%	Weight percent	Concentration
µg/g	Micrograms per gram	Adsorption/Retention
%	Percent	Ratio (e.g., water cut, recovery factor)

References

Cheng, J., Zhou, W., Wang, Q., Cao, G., Bai, W., Zhao, C., & Luo, M. (2014). Technical Breakthrough in Production Engineering Ensures Economic Development of ASP Flooding in Daqing Oilfield. *Society of Petroleum Engineers - SPE Asia Pacific Oil and Gas Conference and Exhibition, APOGCE 2014 - Changing the Game: Opportunities, Challenges and Solutions*, **2**, 941–950. https://doi.org/10.2118/171506-MS

Dean, E., Pitts, M., Wyatt, K., James, D., Mills, K., Al-Murayri, M., & Al-Kharji, A. (2020). Practical Chemical Formulation Design – Time to Break Away from Micellar Polymer Floods, Again. *Proceedings - SPE Symposium on Improved Oil Recovery, 2020-August*. https://doi.org/10.2118/200385-MS

Diab, W. N., & Al-Shalabi, E. W. (2019). Recent Developments in Polymer Flooding for Carbonate Reservoirs under Harsh Conditions. Offshore Technology Conference Brasil 2019, OTCB 2019. https://doi.org/10.4043/29739-MS

Finol, J., Al-Harthy, S., Jaspers, H., Batrani, A., Al-Hadhrami, H., Van Wunnik, J., Stoll, M., Faber, R., & De Kruijf, A. (2012). Alkali-Surfactant-Polymer Flooding Pilot Test in Southern Oman. *Society of Petroleum Engineers - SPE EOR Conference at Oil and Gas West Asia 2012, OGWA - EOR: Building Towards Sustainable Growth*, **2**, 674–685. https://doi.org/10.2118/155403-MS

Gaillard, N., Giovannetti, B., & Favero, C. (2010). Improved Oil Recovery using Thermally and Chemically Protected Compositions Based on co- and ter-polymers Containing Acrylamide. *SPE - DOE Improved Oil Recovery Symposium Proceedings*, **1**, 629–639. https://doi.org/10.2118/129756-MS

Gaillard, N., Giovannetti, B., Leblanc, T., Thomas, A., Braun, O., & Favero, C. (2015). *Selection of Customized Polymers to Enhance Oil Recovery from High Temperature Reservoirs*. https://doi.org/10.2118/177073-MS

Gang, C., He, L., Guochen, S., Guoqing, W., Zhongwen, X., Huaifeng, R., & Ying, Z. (2007). Technical Breakthrough in PCPs' Scaling Issue of ASP Flooding in Daqing Oil Field. *Society of Petroleum Engineers - Progressing Cavity Pumps Conference 2008*, 1–5. https://doi.org/10.2118/109165-MS

Gao, S., He, Y., Zhu, Y., Han, P., Peng, S., & Liu, X. (2020). *Associated Polymer ASP Flooding Scheme Optimization Design and Field Test after Polymer Flooding in Daqing Oilfield*. Proceedings - SPE Symposium on Improved Oil Recovery, 2020-August. https://doi.org/10.2118/200443-MS

Guo, H., Li, Y., Li, Y., Kong, D., Li, B., & Wang, F. (2017). Lessons Learned From ASP Flooding Tests in China. Society of Petroleum Engineers - SPE Reservoir Characterisation and Simulation Conference and Exhibition, RCSC 2017, 226–248. https://doi.org/10.2118/186036-MS

Imanbayev, B. A., Kushekov, R. M., Sagyndikov, M. S., & Shyrakbayev, D. A. (2022, November 15). Feasibility Study of a Polymer Flood for the Uzen Brownfield Conditions. SPE Annual Caspian Technical Conference. https://doi.org/10.2118/212091-MS

Jain, S., Prasad, D., Pandey, A., Koduru, N., & Raj, R. (2020). *Bhagyam ASP Pilot: Successful Formulation Development to Pilot Design and Planning*. Proceedings - SPE Symposium on Improved Oil Recovery, 2020-August. https://doi.org/10.2118/200408-MS

Jiecheng, C., Dianping, X., & Wenguang, B. (2008). Commercial ASP Flood Test in Daqing Oil Field. Society of Petroleum Engineers - 13th Abu Dhabi International Petroleum Exhibition and Conference, ADIPEC 2008, **2**, 654–669. https://doi.org/10.2118/117824-MS

Jouenne, S., Klimenko, A., & Levitt, D. (2017). Polymer Flooding: Establishing Specifications for Dissolved Oxygen and Iron in Injection Water. *SPE Journal*, **22**(02), 438–446. https://doi.org/10.2118/179614-PA

"KMG Engineering" LLP. (2022). *Uzen Field Development Plan* (Vol. **1**).

Levitt, D. B., & Pope, G. A. (2008). Selection and Screening of Polymers for Enhanced-Oil Recovery. *Proceedings - SPE Symposium on Improved Oil Recovery*, **3**, 1125–1142. https://doi.org/10.2118/113845-MS

Levitt, D. B., Slaughter, W., Pope, G. A., & Jouenne, S. (2011). The Effect of Redox Potential and Metal Solubility on Oxidative Polymer Degradation. *SPE Reservoir Evaluation & Engineering*, **14**(03), 287–298. https://doi.org/10.2118/129890-PA

Moradi-Araghi, A., & Doe, P. H. (1987). Hydrolysis and Precipitation of Polyacrylamides in Hard Brines at Elevated Temperatures. *SPE Reservoir Engineering*, **2**(02), 189–198. https://doi.org/10.2118/13033-PA

Ning, S., Barnes, J., Edwards, R., Schulpen, W., Dandekar, A., Zhang, Y., Cercone, D., & Ciferno, J. (2020). *First Ever Polymer Flood Field Pilot to Enhance the Recovery of Heavy Oils on Alaska North Slope – Producer Responses and Operational Lessons Learned*. Proceedings - SPE Annual Technical Conference and Exhibition, 2020-October. https://doi.org/10.2118/201279-MS

Pandey, A., Koduru, N., Stanley, M., Pope, G. A., & Weerasooriya, U. P. (2016). *Results of ASP Pilot in Mangala Field: A Success Story*. Proceedings - SPE Symposium on Improved Oil Recovery, 2016-January. https://doi.org/10.2118/179700-MS

Sagyndikov, M. S., Kushekov, R. M., & Seright, R. S. (2022). Review of Important Aspects and Performances of Polymer Flooding versus ASP Flooding. *Bulletin of the Karaganda University. "Chemistry" Series*, **107**(3), 35–55. https://doi.org/10.31489/2022Ch3/3-22-13

Sagyndikov, M. S., Salimgarayev, I. I., Ogay, E. K., Seright, R. S., & Kudaibergenov, S. E. (2022). Assessing polyacrylamide solution chemical stability during a polymer flood in the Kalamkas field, Western Kazakhstan. *Bulletin of the Karaganda University. "Chemistry" Series*, **105**(1), 99–112. https://doi.org/10.31489/2022Ch1/99-112

Sagyndikov, M. S., Seright, R. S., Kudaibergenov, S., & Ogay, E. (2022). Field Demonstration of the Impact of Fractures on Hydrolyzed Polyacrylamide Injectivity, Propagation, and Degradation. *SPE Journal*, **27**(02), 999–1016. https://doi.org/10.2118/208611-PA

Seright, R. S. (2017). How Much Polymer Should Be Injected During a Polymer Flood? Review of Previous and Current Practices. *SPE Journal*, **22**(01), 1–18. https://doi.org/10.2118/179543-PA

Seright, R. S., Campbell, A. R., Mozley, P. S., & Han, P. (2010). Stability of Partially Hydrolyzed Polyacrylamides at Elevated Temperatures in the Absence of Divalent Cations. *SPE Journal*, **15**(02), 341–348. https://doi.org/10.2118/121460-PA

Seright, R. S., & Skjevrak, I. (2015). Effect of Dissolved Iron and Oxygen on Stability of Hydrolyzed Polyacrylamide Polymers. *SPE Journal*, **20**(03), 433–441. https://doi.org/10.2118/169030-PA

Seright, R. S., Wavrik, K. E., Zhang, G., & AlSofi, A. M. (2021). Stability and Behavior in Carbonate Cores for New Enhanced-Oil-Recovery Polymers at Elevated Temperatures in Hard Saline Brines. *SPE Reservoir Evaluation & Engineering*, **24**(01), 1–18. https://doi.org/10.2118/200324-PA

Shankar, V., Zagitov, R., Shekhar, S., Gupta, A. K., Kumar, M. S., Kumar, R., Veerbhadrappa, S., & Nakutnyy, P. (2023). Evaluation of ATBS Polymers for Mangala Polymer Flood. *SPE Reservoir Evaluation & Engineering*, **26**(03), 722–736. https://doi.org/10.2118/211461-PA

Sipachev N.V., & Posevich A.G. (1981). Characteristics of the Displacement of Oil by Water (in Russian). *Nedra*.

Song, H., Ghosh, P., Mejia, M., & Mohanty, K. K. (2022). Polymer Transport in Low-Permeability Carbonate Rocks. *SPE Reservoir Evaluation & Engineering*, **25**(04), 900–913. https://doi.org/10.2118/206024-PA

Volokitin, Y., Shuster, M., Karpan, V., Mikhaylenko, E., Koltsov, I., Rakitin, A., Tkachev, I., & Podberezhny Salym, M. (2017). *West Salym ASP Pilot: Surveillance Results and Operational Challenges*. Society of Petroleum Engineers - SPE Russian Petroleum Technology Conference 2017. https://doi.org/10.2118/187838-MS

Wang, D., Dong, H., Lv, C., Fu, X., & Nie, J. (2009). Review of Practical Experience by Polymer Flooding at Daqing. *SPE Reservoir Evaluation & Engineering*, **12**(03), 470–476. https://doi.org/10.2118/114342-PA

Wang, D., Seright, R. S., Shao, Z., & Wang, J. (2008). Key Aspects of Project Design for Polymer Flooding at the Daqing Oilfield. *SPE Reservoir Evaluation & Engineering*, **11**(06), 1117–1124. https://doi.org/10.2118/109682-PA

Wellington, S. L. (1983). Biopolymer Solution Viscosity Stabilization—Polymer Degradation and Antioxidant Use. *Society of Petroleum Engineers Journal*, **23**(06), 901–912. https://doi.org/10.2118/9296-PA

Zhao, Y., Yin, S., Seright, R. S., Ning, S., Zhang, Y., & Bai, B. (2020). *Performance of Low Salinity Polymer Flood in Enhancing Heavy Oil Recovery on the Alaska North Slope*. https://doi.org/10.15530/URTEC-2020-1082